Asset Management Module — Complete Documentation
Module: `asset_management` | **Version:** 1.0 | **Author:** Hagbes Plc

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1. Prerequisites & Hardware Requirements

Software Prerequisites
Requirement	Minimum	Recommended
Odoo Version	17.0	17.0 latest
Python	3.10	3.11+
PostgreSQL	14	15+
Browser	Chrome 110+	Chrome/Edge latest

Required Odoo Dependencies
Module	Purpose
`base`, `hr`, `stock`, `account`	Core framework
`om_account_asset`	Accounting depreciation
`approval_central`	Multi-step approval workflows
`image_capture_upload_widget`	In-app photo capture
`workshop_management`	Vehicle asset inspection

Python Libraries
```bash
pip install qrcode pillow
```

Hardware Requirements

Office / Desktop
Device	Minimum Spec
Computer / Laptop	Intel Core i3, 4 GB RAM
Monitor	1366×768 resolution
Internet	5 Mbps stable

Field Auditors (Recommended)
Device	Minimum Spec	Purpose
Tablet (Android 10+ / iPad iOS 14+)	8" screen, 3 GB RAM	QR scanning, on-site audit
Smartphone (Android 10+)	Rear camera 8 MP+	QR code scanning
Barcode / QR Scanner (USB or Bluetooth)	Standard HID scanner	Warehouse scanning

> **Important:** QR scanning requires a working camera or barcode scanner. The module uses the browser camera API or HID keyboard input.

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2. Installation Guide

Step 1 — Copy module
```bash
cp -r asset_management /path/to/odoo/custom_addons/
```

Step 2 — Install dependencies (in order)
Apps → Search and install:
1. `om_account_asset`
2. `approval_central`
3. `image_capture_upload_widget`

4. `workshop_management`

Step 3 — Install Asset Management
Apps → Search `Asset Management with QR` → **Install**

Step 4 — Post-Installation Checklist
- [ ] Set company **Prefix** (Settings → Companies → Prefix field)
- [ ] Configure **Analytic Plans** for Branches
- [ ] Set up **Approval Flows** for `am.asset.handover.request` and `am.asset.audit`
- [ ] Configure **Asset Categories** (Major & Minor)
- [ ] Configure **Warehouses & Locations**
- [ ] Verify scheduled action `_cron_generate_depreciation` is active

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3. Access & User Roles

Role Summary

Role	Technical Name	Description
User	`group_asset_user`	View assigned assets, submit handover requests
Property Manager	`group_asset_property_manager`	Assign assets to any employee or warehouse
Manager	`group_asset_manager`	Full access — categories, batches, approvals
Technical Inspector	`group_asset_inspector`	Conduct and submit asset audits
Financial Auditor	`group_asset_financial_auditor`	View and post depreciation journal entries

> **Hierarchy:** Manager ⊃ Property Manager ⊃ User

Permission Matrix

Action	User	Prop. Manager	Manager	Inspector	Fin. Auditor
View own assets	✅	✅	✅	✅	✅
Create asset	❌	✅	✅	❌	❌
Assign to employee	Own reports	Any	Any	❌	❌
Assign to warehouse	❌	✅	✅	❌	❌
Conduct audit	❌	❌	✅	✅	❌
Post depreciation	❌	❌	✅	❌	✅
Configure categories	❌	❌	✅	❌	❌

Assigning Roles
Settings → Users & Companies → Users → Open user → Under **"Asset Management"** section → Select role → Save

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4. User Manual

4.1 Dashboard
Navigation: Asset Management → Dashboard

Real-time overview showing: Total Assets, Net Book Value, Warehouse vs Employee location counts, Asset by Category, By Branch (Top 5), Status breakdown (Draft/Running/Closed), Overdue & Upcoming Audits.

4.2 Asset Registration
Navigation: Asset Management → Assets

Creating a New Asset

Field	Description	Required
Asset Name	Descriptive name	✅
Employee Name	Responsible employee	✅
Major Category	Top-level category	✅
Minor Category	Sub-category	Optional
Branch	Analytic account / branch	✅
Acquisition Date	Date of purchase	Required to validate
Original Cost	Purchase cost (must be > 0)	Required to validate
Serial Number	Physical serial	Optional
Audit Frequency	Monthly / Quarterly / Yearly / Never	Optional

Auto-generated on save:
- QR Code Number: `COMPANY-BRANCH-MAJOR-MINOR-XXXX`
- QR Code image (PNG)
- Specification lines from minor category template

For Vehicle Assets: Enable "Is Vehicle Asset" → fill Vehicle Details tab (mandatory).
For Component Assets: Enable "Is Component" → link to a Parent Asset.

Asset States
State	Description
Draft	Initial — editable
Running	Validated and active
Closed	Disposed / decommissioned

Click **Validate** to move Draft → Running (requires Acquisition Date + Cost > 0).

4.3 Asset Categories
Navigation: Asset Management → Configuration → Major / Minor Categories

Major Category — Key Fields
Field	Purpose
Code	Used in QR number
Depreciation Rate (%)	Annual depreciation
Accounting Category	Links to `om_account_asset`
Journal, Asset Account, Depreciation Accounts	For posting entries

Minor Category — Key Fields
Field	Purpose
Code	Used in QR number
Audit Frequency	Default for assets in this category
Audit Configuration Toggles	Which audit sections are required
Specification Templates	Auto-added specs to new assets

4.4 Asset Handover Request
Navigation: Asset Management → Handover → Handover Requests

Workflow
```
```

```
Draft → Pending Approval → Awaiting Pre-Transfer Audit → Audit Done → Approved / Rejected
```

Steps:
1. Click New → Select Assets (from your assigned assets)
2. Select "To Employee" or "To Warehouse"
3. Click **Submit for Approval**
4. Approval flow triggers → at Audit Step, one pre-transfer audit is created per asset
5. All audits must reach "Approved" state
6. Approver clicks **Approve** → assets are transferred automatically

Who can assign to warehouse? Only Property Manager or Manager.
Who can assign to any employee? Only Property Manager or Manager.
Standard users can only assign to their direct/indirect subordinates.

4.5 Asset Audit
Navigation: Asset Management → Audits

Audit Types
Type	Description
Simple Scan	Scan QR to confirm presence
Specification Check	Verify specs match records
Vehicle Inspection	Full vehicle checklist
Custom Configured	From minor category settings

Audit Workflow
```
Draft → In Progress → Submitted → Approved (Done) / Rejected
```

Conducting an Audit:
1. Open audit → Click **Start**
2. Fill required sections:
   - **QR Scan:** Place cursor in scan field → scan QR → green confirmation
   - **Specs:** Verify actual vs expected values
   - **Vehicle:** Fill exterior/interior/mechanical checklists
   - **Photos:** Capture if required
3. Set **Final Recommendation**
4. Click **Submit** → approval flow
5. Approver clicks **Approve**

> **Frequency Gate:** Cannot create a manual audit before the next audit date is reached. Batch/handover/transfer audits bypass this gate.

4.6 Audit Batch
Navigation: Asset Management → Audits → Audit Batches

1. Click New → Set Batch Name, Audit Date, Branch filter
2. Click **Generate Audits** → creates one audit per qualifying asset
3. Auditors conduct each audit individually

4.7 Warehouse & Location Management
Navigation: Asset Management → Configuration → Warehouses / Locations

- **Warehouses** — Physical storage for unassigned assets

- **Locations** — Sub-locations within a warehouse

> Direct creation/deletion of stock records is blocked. All movements must go through Handover Requests or the Transaction Wizard.

4.8 Temporary Asset Requests
Navigation: Asset Management → Requests → Temporary Requests

For short-term borrowing. Includes request date range, return tracking, and approval flow.

4.9 Depreciation
Navigation: Open Asset → Depreciation History tab

Auto-generated yearly depreciation lines based on:
- Depreciation Rate on Major Category
- Original Cost
- Acquisition Date

Formula: `Yearly Amount = Original Cost × (Rate / 100)`
Capped so total never exceeds Original Cost.

To post a depreciation entry: Open asset → Depreciation History tab → Click **Post** on a line → Journal entry is created and posted.

4.10 QR Code Scanning

QR format: `COMPANY-BRANCH-MAJOR-MINOR-XXXX` (e.g., `HGB-AA-IT-LAP-0001`)

During Audit:
- Open audit form → place cursor in "Scan QR Code" field
- **Tablet/Phone:** Camera launches automatically
- **USB/BT Scanner:** Scan directly into the field
- System validates against asset's QR number and URL

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5. Configuration

Approval Flows
Go to **Approval Central → Flows** and create flows for:
- `am.asset.handover.request` — mark one step as **"Is Audit Step"**
- `am.asset.audit`

Company Prefix
Settings → Companies → [Company] → **Prefix** field (e.g., `HGB`)

Scheduled Actions
Verify active: Settings → Technical → Automation → Scheduled Actions
`am.asset._cron_generate_depreciation` — should run **Yearly**

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6. Reports

Report	Access	Description
Asset Audit Report	Audit form → Print	Full audit checklist PDF

| Asset Handover Report | Handover form → Print | Handover certificate with signature area |

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7. Troubleshooting

Problem	Cause	Solution
QR not generated	Missing company/branch/category	Fill all 3 fields before saving
Cannot validate	Missing date or cost ≤ 0	Fill both required fields
Handover stuck	Pre-transfer audits not approved	Complete all linked audits
Cannot create audit	Next audit date not reached	Wait for due date or use batch route
Scan not recognized	Wrong QR format	Scan QR printed from system only
Depreciation not posting	Missing accounts on category	Configure accounts on Major Category
Cannot assign to warehouse	Not a Property Manager	Request Property Manager to perform